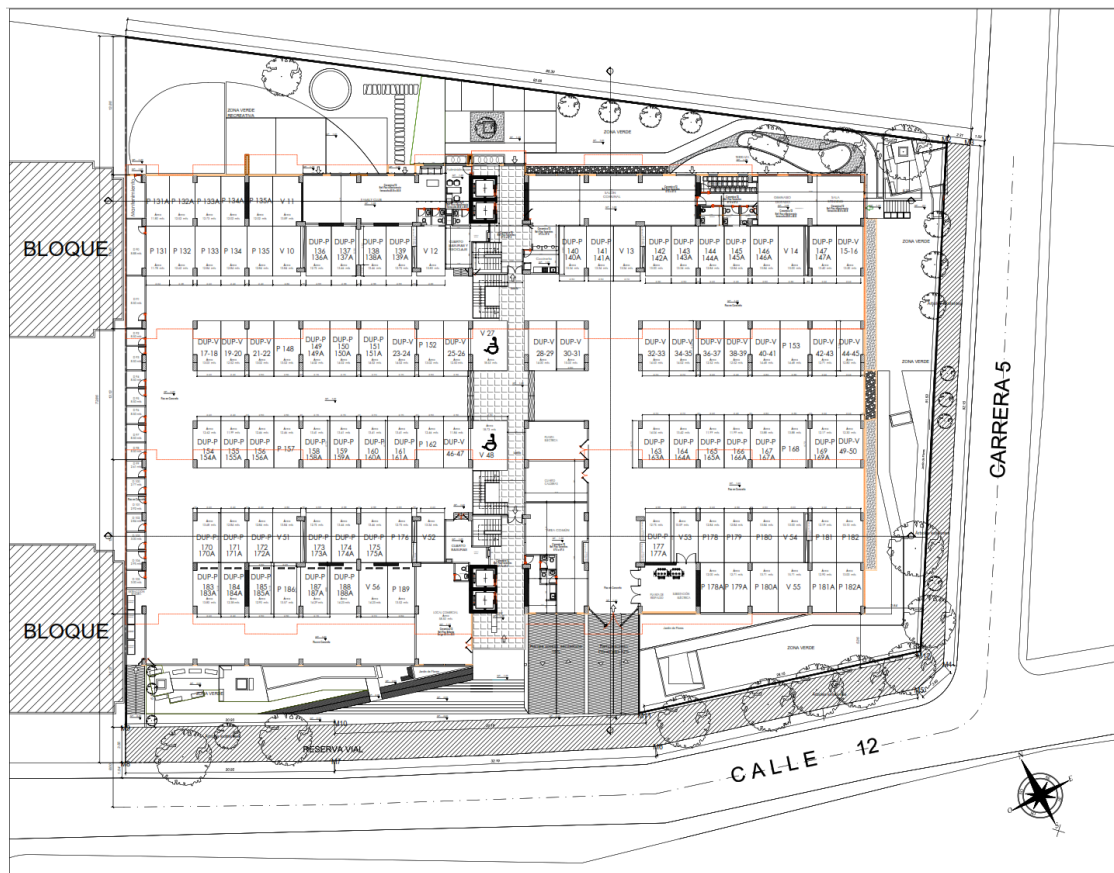


Aranjuez Building Spatial Network Analysis

Aranjuez is an apartment building with two towers of eight floors, one basement, and one access level facing the street. This project focuses on the analysis of this access level using a Spatial Network Analysis approach, because this floor has an important mix of uses and flows.

The analyzed level includes common areas, pedestrian entrances, vehicular entrances, parking spaces, service areas, and stairs. This makes it an important space to study, because pedestrian circulation, vehicular circulation, entrance points, transition areas, and vertical connections to the towers happen in the same level.

Through graph analysis, this project aims to identify patterns of connectivity, accessibility, spatial hierarchy, and possible critical points in the circulation system. The results can help identify important topics for future design decisions, management, or architectural interventions, based on the spatial relationships shown by the graph structure.



1. Degree Centrality to identify which spaces have more direct connections.

1. Connectivity Heatmap

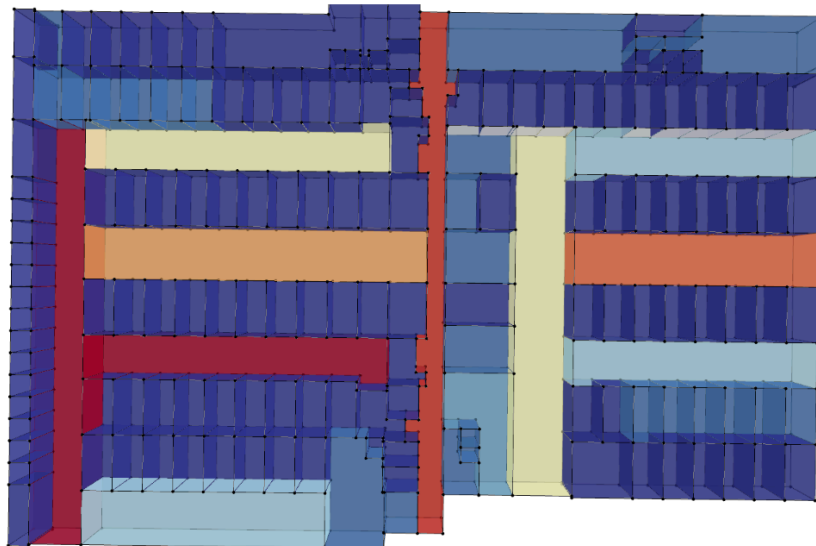
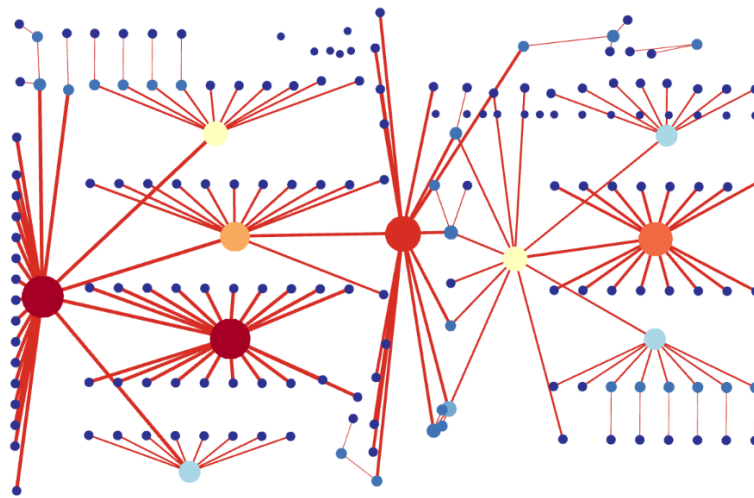
Degree Centrality

Blue = low | Yellow/Orange = medium | Red = high

Node size follows the same metric value.

Degree Centrality / Connectivity

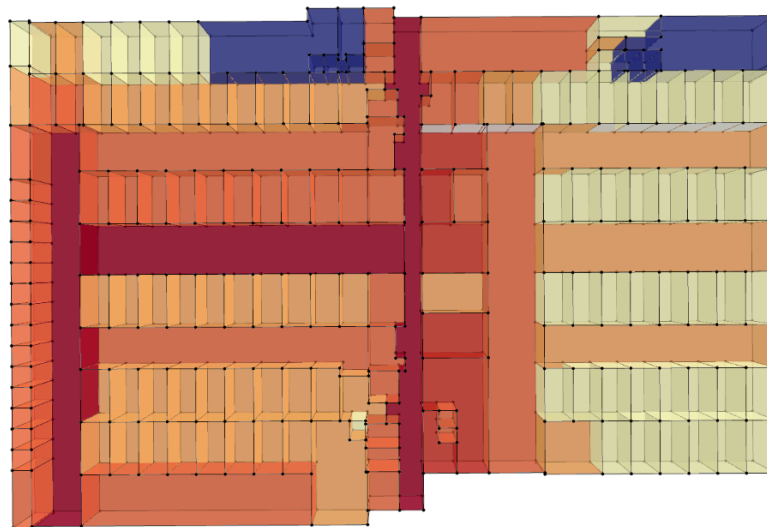
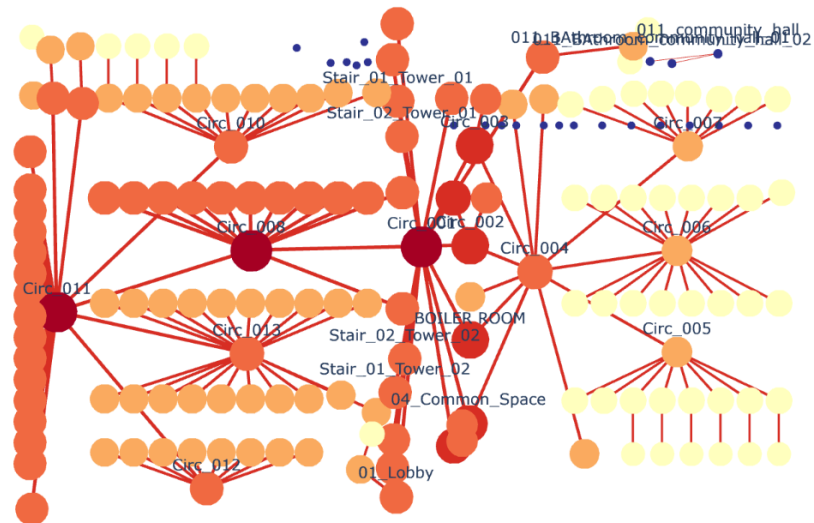
Vertices: 177 Edges: 162



- Most directly connected space: Circulation_011 with degree centrality 0.125.

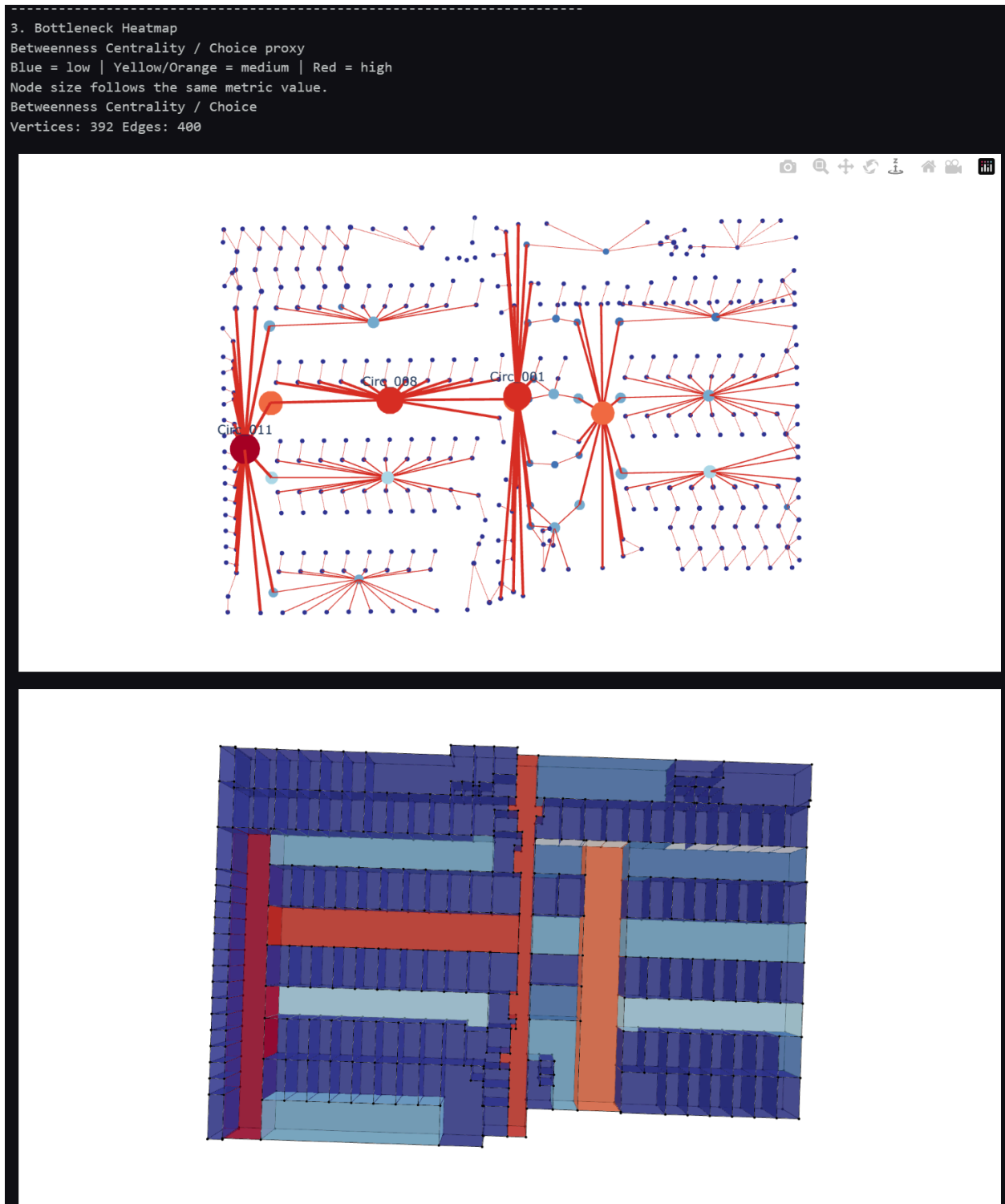
2.Closeness Centrality to find which spaces are “closer” to the rest in topological terms.

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2. Accessibility Heatmap
Closeness Centrality / Integration proxy
Blue = low | Yellow/Orange = medium | Red = high
Node size follows the same metric value.
Closeness Centrality / Integration
Vertices: 177 Edges: 162
```



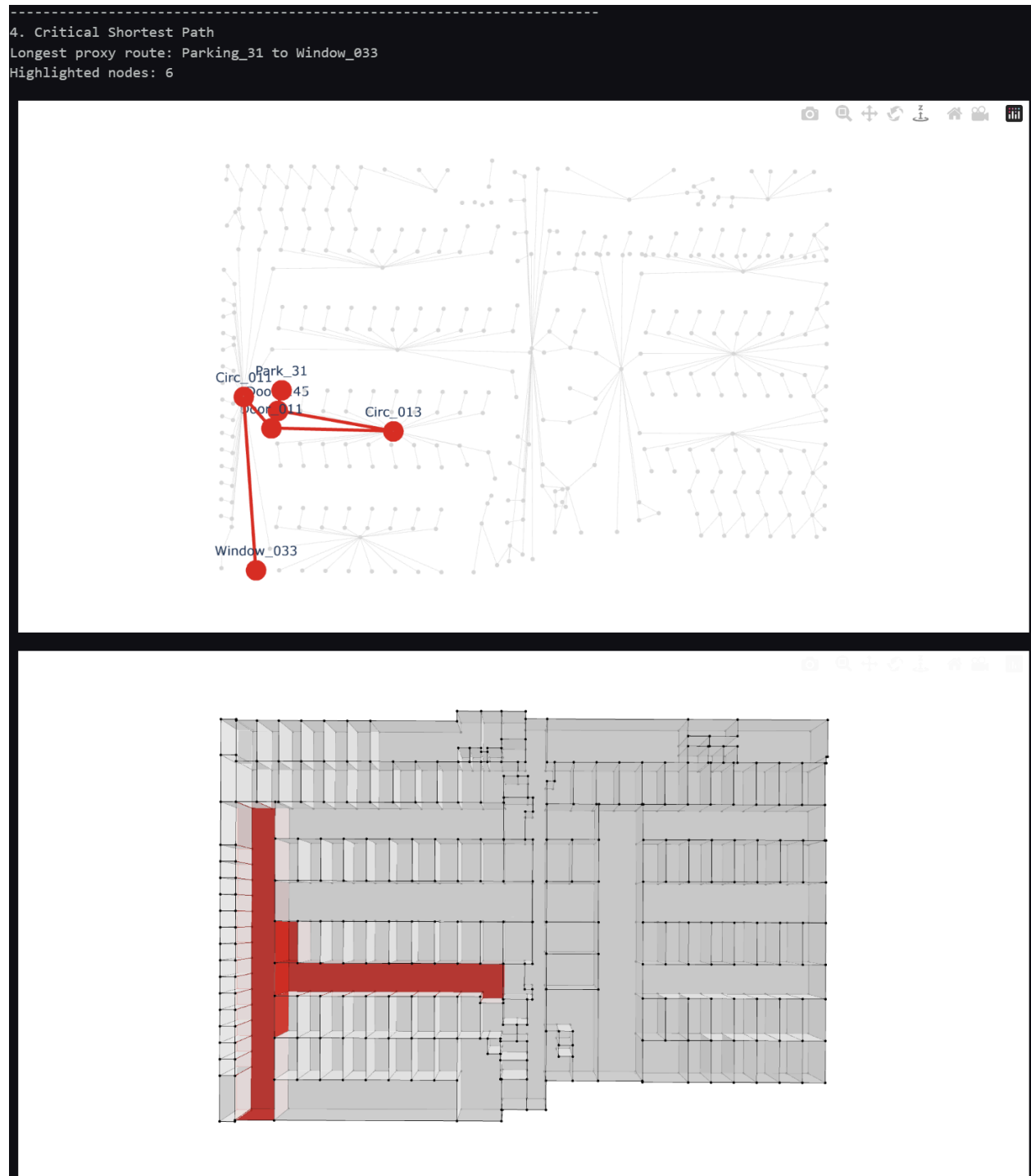
- Most globally accessible space: Circulation_008 with closeness centrality 1.0.

4. Betweenness Centrality to detect critical connectors and bottlenecks.



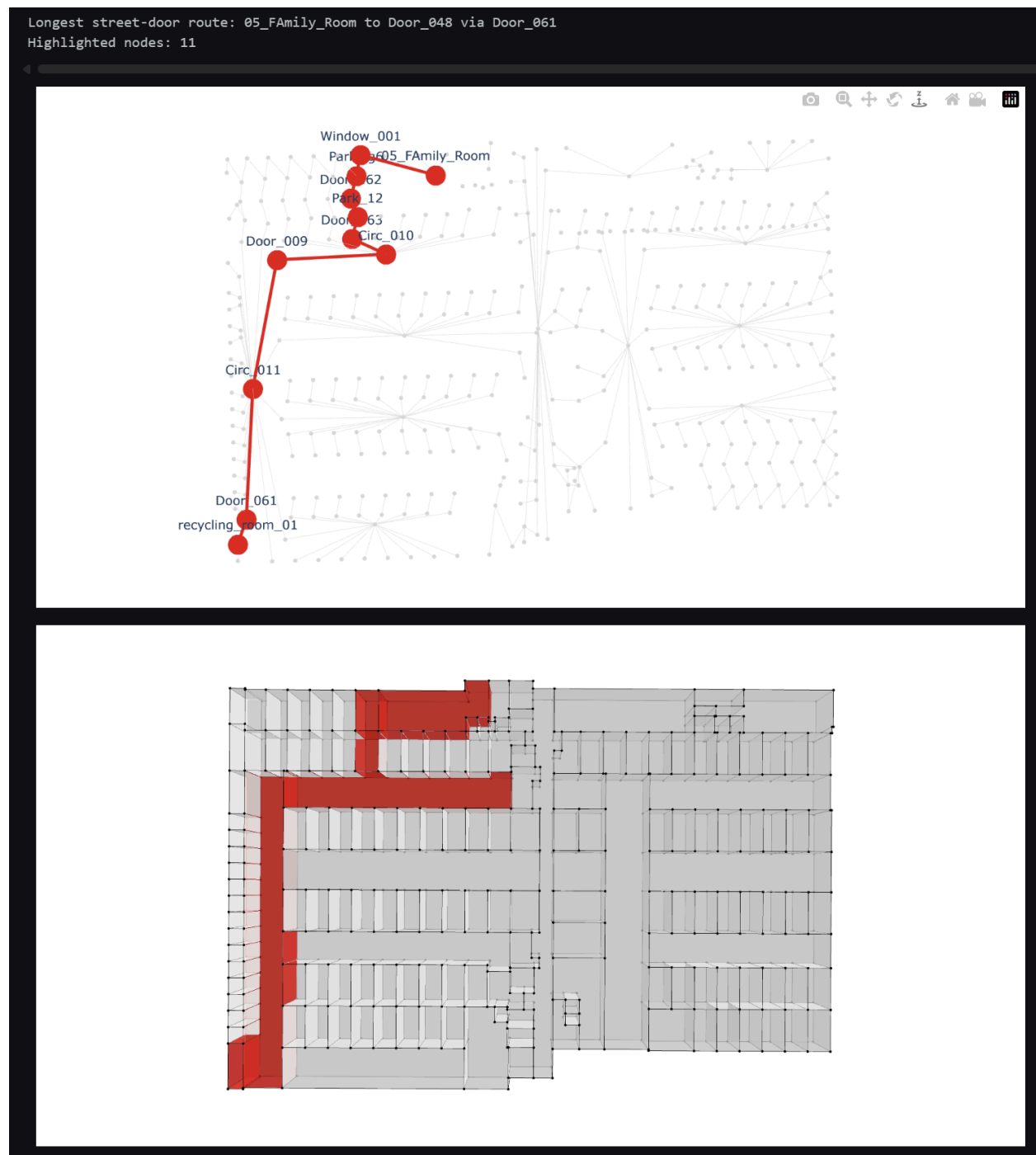
- Strongest bottleneck / connector: Circulation_011 with betweenness centrality 0.598224.

5. Shortest Paths to estimate more efficient routes within the network.



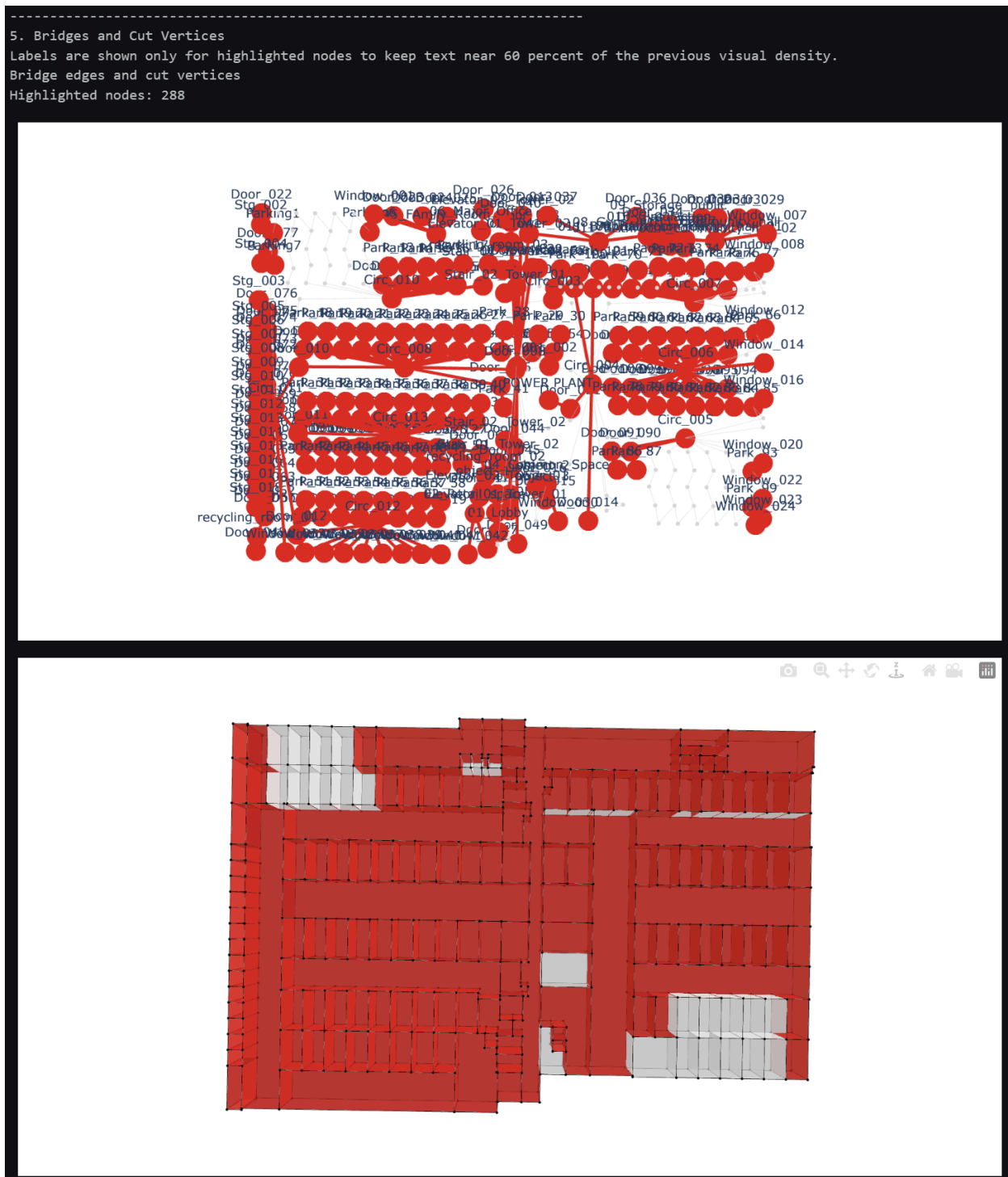
- Longest proxy route to an exterior aperture starts at Parking_31 and reaches Window_033 in 59.295304 model units.

5a. Shortest Paths to estimate more efficient routes within the network.



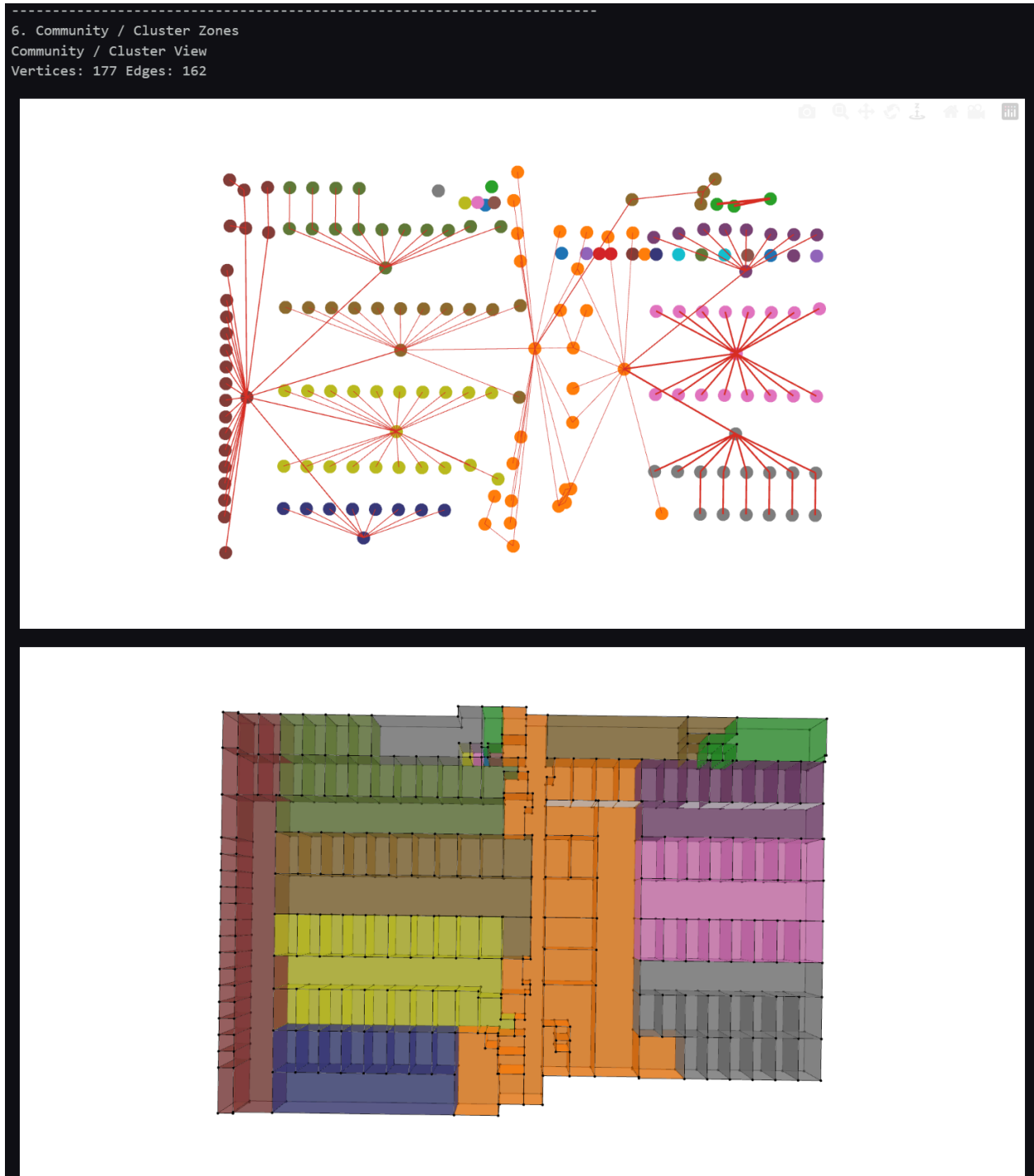
- This result shows the longest route from an interior space to a street exit. The route starts in 05_Family_Room and goes to Door_048, but it must pass first through Door_061.

5b. Bridge edges are critical connections. If one of these connections is removed or blocked, part of the system may become disconnected.



- The result shows 288 highlighted nodes, which means there are many nodes related to critical connections or cut-off points. This may indicate that the network relies on many specific points to maintain continuity of movement.

6. Community / Cluster Zones A community is a group of nodes that have more connections among themselves than with the rest of the system.



- This result shows that the access level is not one uniform system. It is organized into different spatial zones. These zones may be related to parking areas, circulation spaces, services, entrances, or vertical connections to the towers. This analysis helps us understand how the building is divided into groups and how these groups are connected.

7. Conclusions

- The results show that Circulation_011 is the most connected node and also a critical connector, while Circulation_008 has the highest overall accessibility.
- Additionally, doors with high between-use traffic, acting as bottlenecks, were detected.
- Community detection confirmed the existence of functional zones within the building.
- Thus, graph analysis proves to be a powerful tool for objectively interpreting circulation, spatial hierarchy, and connectivity in architectural datasets.